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TITLE PAGE

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Problem Statement ID – SIH26165

Problem Statement Title – AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports

Theme – Smart Automation

PS Category – Software

Team ID – ____

Team Name – Signal-0

SANKET
SIF Precursor Detection

Reads a free-text safety report and answers one question: could this have killed someone?

Three gates · hazard, control status, plausible severity · tagged to the IOGP Life-Saving Rules

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SANKET — FROM INCIDENT NARRATIVES TO EARLY SAFETY WARNINGS

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SANKET analyzes safety and near-miss reports using AI to detect potential Serious Injury & Fatality (SIF) precursors.

IDENTIFIES

- Hazard & Life-Saving Rule
- Control status
- Severity (1–5)
- SIF Precursor: Yes / No
- Evidence & reasoning

INNOVATION

- LLM + ML hybrid approach
- Explainable risk assessment
- One-Change severity rule
- Prioritized safety-review queue

THE DECISION, MADE VISIBLE

GATE 1
Hazard present?
yes / no / insufficient

→

GATE 2
Was a control doing its job?
absent / failed / present / unclear

→

GATE 3
Plausible worst case?
severity 1–5

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TECHNICAL APPROACH

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TECH

- React
- FastAPI · Python
- Groq · GPT-OSS-20B
- TF-IDF + Logistic Regression
- Supabase (Postgres)

SQLite is the on-instance classification cache only.

FLOW

1. **Report**
2. AI / NLP
3. Hazard
4. Life-Saving Rule
5. Control status
6. Severity
7. SIF Decision
8. **Risk Priority**

SIF RULE

Hazard = YES

+ Control = Absent / Failed

+ Severity ≥ 4

→ SIF PRECURSOR

Computed from the gates, never typed. A database CHECK constraint enforces it, so no model can emit a contradicting row.

DESIGNED TO DEGRADE, NOT TO FAIL

Every call runs under a hard deadline. A rate-limited request retries inside that deadline, then a locally trained TF-IDF model answers instead — and the response says so. The API reports which classifier actually served it, so a degraded answer is visible rather than silent.

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FEASIBILITY AND VIABILITY

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FEASIBLE BECAUSE

- Uses existing incident reports
- No specialized hardware
- API-based & scalable
- LLM + local fallback
- Human-in-the-loop

CHALLENGES → SOLUTIONS

- Unstructured text → LLM analysis
- Ambiguous controls → 4-state classification
- API failure → retry inside a deadline, then fallback
- Domain variation → tested on 30 real OSHA narratives (F1 0.118)

THE CONSTRAINT WE ACTUALLY HIT

Not compute — tokens. The free tier caps us near 63 reports a day, so a full re-classification takes days, not minutes. Everything downstream is built around that: the classifier is resumable, refuses to store a fallback answer as though it were real, and the dashboard keeps reporting on the last model version that covers the corpus instead of going blank mid-migration.

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IMPACT AND BENEFITS

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POTENTIAL IMPACT

- Early detection of SIF precursors in OIL safety reports
- Prioritized review of high-risk incidents
- Ranks sites by precursor RATE, not count — so a site is not punished for reporting diligently

BENEFITS

- Safety: proactive intervention before serious incidents
- Operational: removes the manual first-pass screen
- Economic: prevents costly injuries, incidents and downtime
- Scalable: ten reports or ten thousand, same pipeline

Safety Reports → SIF Precursors → Risk Prioritization → Early Action → Safer Operations

It never closes a report. It reorders the queue a human already reads, and puts the ones with fatal potential at the top.

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RESEARCH AND REFERENCES

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Research:

IOGP Life-Saving Rules · DEKRA SIF Framework · EEI SIF Model

Data:

150 synthetic reports + 30 real OSHA severe-injury narratives · 173 gold labels

PROTOTYPE EVALUATION

HUMAN CEILING

0.947 Cohen's kappa

97.4% agreement · two annotators, independently · n=38

TF-IDF BASELINE

0.754 F1 ± 0.077

5-fold cross-validated · PR-AUC 0.847 · n=114

OSHA GENERALISATION

0.118 F1

real narratives nobody on the team wrote · n=30, too small to conclude from

LLM figure withdrawn, deliberately — we found a held-out report inside our own prompt and pulled the number rather than ship it. Re-measurement in progress.

STATED UP FRONT, BECAUSE A JUDGE WILL FIND THEM

Our precursor rate is 58.7%, not the 20–25% the problem statement cites — our generator centred every synthetic report on a hazard. The ceiling is a 38-report re-label, so we quote the subset. We report no accuracy figure, and never a baseline scored on its own training data.

GitHub: SANKET — AI-powered SIF precursor detection